As usual, let $B^\top$ denote the transpose of B. Then B and $B^\top$ define linear maps $B\colon \mathcal{E}(G) \to \mathcal{V}(G)$ and $B^\top\colon \mathcal{V}(G) \to \mathcal{E}(G)$ with respect to the standard bases. As is easy to check, B maps an edge set $F \subseteq E$ to the set of vertices incident with an odd number of edges in F, while $B^\top$ maps a set $U \subseteq V$ to set of edges with exactly one end in U. In particular:

Proposition 1.9.6.

(i) *The kernel of B is $\mathcal{C}(G)$.*

(ii) *The image of $B^\top$ is $\mathcal{B}(G)$.* □

More on this in the exercises and notes at the end of this chapter.

The *adjacency matrix* $A = (a_{ij})_{n\times n}$ of G is defined by *adjacency matrix*

$$a_{ij} := \begin{cases} 1 & \text{if } v_iv_j \in E \\ 0 & \text{otherwise.} \end{cases}$$

Viewed as a linear map $\mathcal{V} \to \mathcal{V}$, the adjacency matrix maps a given set $U \subseteq V$ to the set of vertices with an odd number of neighbours in U.

Let D denote the real diagonal matrix $(d_{ij})_{n\times n}$ with $d_{ii} = d(v_i)$ and $d_{ij} = 0$ otherwise. Our last proposition establishes a connection between A and B (now viewed as real matrices), which can be verified simply from the definition of matrix multiplication:

Proposition 1.9.7. *$BB^\top = A + D$.* □

It is also instructive to check that $A + D$, with entries taken mod 2, defines the same map $\mathcal{V} \to \mathcal{V}$ as the composition of the maps of B and $B^\top$ (Exercise 57).

1.10 Other notions of graphs

For completeness, we now mention a few other notions of graphs which feature less frequently or not at all in this book.

A *hypergraph* is a pair (V, E) of disjoint sets, where the elements of E are non-empty subsets (of any cardinality) of V. Thus, graphs are special hypergraphs. *hypergraph*

A *directed graph* (or *digraph*) is a pair (V, E) of disjoint sets (of *vertices* and *edges*) together with two maps $\text{init}\colon E \to V$ and $\text{ter}\colon E \to V$ assigning to every edge e an *initial vertex* $\text{init}(e)$ and a *terminal vertex* $\text{ter}(e)$. The edge e is said to be *directed from* $\text{init}(e)$ *to* $\text{ter}(e)$. Note that a directed graph may have several edges between the same two vertices x, y. Such edges are called *multiple edges*; if they have the same direction (say from x to y), they are *parallel*. If $\text{init}(e) = \text{ter}(e)$, the edge e is called a *loop*. *directed graph* $\text{init}(e)$ $\text{ter}(e)$ *loop* orientation

oriented graph

A directed graph D is an *orientation* of an (undirected) graph G if $V(D) = V(G)$ and $E(D) = E(G)$, and if $\{\text{init}(e), \text{ter}(e)\} = \{x, y\}$ for every edge $e = xy$. Intuitively, such an *oriented graph* arises from an undirected graph simply by directing every edge from one of its ends to the other. Put differently, oriented graphs are directed graphs without loops or multiple edges.

multigraph

A *multigraph* is a pair (V, E) of disjoint sets (of *vertices* and *edges*) together with a map $E \to V \cup [V]^2$ assigning to every edge either one or two vertices, its *ends*. Thus, multigraphs too can have loops and multiple edges: we may think of a multigraph as a directed graph whose edge directions have been 'forgotten'. To express that x and y are the ends of an edge e we still write $e = xy$, though this no longer determines e uniquely.

A graph is thus essentially the same as a multigraph without loops or multiple edges. Somewhat surprisingly, proving a graph theorem more generally for multigraphs may, on occasion, simplify the proof. Moreover, there are areas in graph theory (such as plane duality; see Chapters 4.6 and 6.5) where multigraphs arise more naturally than graphs, and where any restriction to the latter would seem artificial and be technically complicated. We shall therefore consider multigraphs in these cases, but without much technical ado: terminology introduced earlier for graphs will be used correspondingly.

A few differences, however, should be pointed out. A multigraph may have cycles of length 1 or 2: loops, and pairs of multiple edges (or *double edges*). A loop at a vertex makes it its own neighbour, and contributes 2 to its degree; in Figure 1.10.1, we thus have $d(v_e) = 6$. The ends of loops and parallel edges in a multigraph G are considered as separating that edge from the rest of G. The vertex v of a loop e, therefore, is a cutvertex unless $(\{v\}, \{e\})$ is a component of G, and $(\{v\}, \{e\})$ is a 'block' in the sense of Chapter 3.1. Thus, a multigraph with a loop is never 2-connected, and any 3-connected multigraph is in fact a graph.

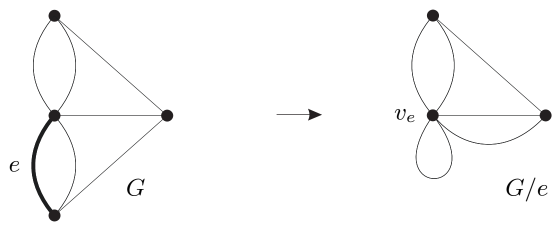

Fig. 1.10.1. Contracting the edge e in the multigraph corresponding to Fig. 1.8.1

The notion of edge contraction is simpler in multigraphs than in graphs. If we contract an edge $e = xy$ in a multigraph $G = (V, E)$ to a new vertex v_e, there is no longer a need to delete any edges other than

e itself: edges parallel to e become loops at v_e, while edges xv and yv become parallel edges between v_e and v (Fig. 1.10.1). Thus, formally, $E(G/e) = E \smallsetminus \{e\}$, and only the incidence map $e' \mapsto \{\text{init}(e'), \text{ter}(e')\}$ of G has to be adjusted to the new vertex set in G/e. Contracting a loop thus has the same effect as deleting it.

The notion of a minor adapts accordingly. The contraction minor G/P defined by a partition P of $V(G)$ into connected sets has precisely those edges of G that join distinct partition classes. If there are several such edges between the same two classes, they become parallel edges of G/P. However, we do not normally give G/P any loops resulting from edges of G whose ends lie in the same partition class U. This would require us to say which of the edges of $G[U]$ are contracted (assuming they induce a connected spanning subgraph of $G[U]$), or at least how many are, which seems futile if we do not care about loops in G/P anyway.

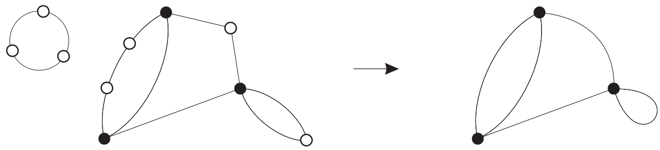

Fig. 1.10.2. Suppressing the white vertices

suppressing a vertex

If v is a vertex of degree 2 in a multigraph G, then by *suppressing* v we mean deleting v and adding an edge between its two neighbours. (If its two incident edges are identical, i.e. form a loop at v, we add no edge and obtain just $G - v$. If they go to the same vertex $w \neq v$, the added edge will be a loop at w. See Figure 1.10.2.) Since the degrees of all vertices other than v remain unchanged when v is suppressed, suppressing several vertices of G always yields a well-defined multigraph that is independent of the order in which those vertices are suppressed.

Finally, it should be pointed out that authors who usually work with multigraphs tend to call them 'graphs'; in their terminology, our graphs would be called 'simple graphs'.

Exercises

1.$^-$ What is the number of edges in a K^n?

2. Let $d \in \mathbb{N}$ and $V := \{0,1\}^d$; thus, V is the set of all 0–1 sequences of length d. The graph on V in which two such sequences form an edge if and only if they differ in exactly one position is called the *d-dimensional cube*. Determine the average degree, number of edges, diameter, girth and circumference of this graph.

   (Hint for the circumference: induction on d.)